Lesson plan · for the teacher

Inside the loop

AI agents · Level 3 · Lesson 1

Time
40 min

Materials
A long strip of paper (or A4 sheets taped together); Blue and green pens or chalk; A request card, such as “What is 6B’s first lesson tomorrow?”, and the class timetable

Learning goals

- Read an agent’s conversation as one text with markers.
- Say who writes each part: the model or the app.
- Explain why every loop needs a stop.

Activity: One long text

1. Warm-up (5 min): Ask: when an agent plans a trip in two steps, where does it keep the first step’s answer? Collect ideas, then reveal: in one long text that grows.
2. Teach (10 min): Write the six markers on the board (tools, user, plan, call, result, answer). Explain that the model writes the plan, the call and the answer, the app writes the rest, and the model stops after a call and waits.
3. One long text (15 min): On a paper strip, the class writes one conversation for “What is 6B’s first lesson tomorrow?”, part by part: model parts in blue, app parts in green. Then add the trip from Kafi to Dala, which needs two calls.
4. Check it (5 min): Ask which colour can make mistakes (blue: the model) and where the app could check a call before the green result is written.
5. Wrap-up (5 min): Ask what would happen if the model never wrote an answer, and agree on a step limit. Then give the exit ticket.

Prepare a long paper strip (or A4 sheets taped together), blue and green pens or chalk, and the class timetable on the board. No phones are needed.

Discussion

- Why is it safer that only the app runs tools and writes results?
- What could go wrong in a loop with no step limit?
- Where in the loop would you put a check for a message to parents?

Exit ticket

- Write the six parts of an agent’s text in order, and circle the ones the model writes.

Try it at school tomorrow
Write one agent conversation on a long paper strip with your class: the model’s parts (plan, call, answer) in blue, the app’s parts (tools line, request, result) in green. Then ask: which colour can make mistakes, and where would the app check?

Practice questions · for you

Inside the loop

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1 **An agent keeps calling the weather tool and never answers. What should the app have?**

◯ A bigger model ◯ A faster phone ◯ A step limit that stops the loop and tells a person

2 **Match each marker to what follows it in Darsi Tiny's text.**

<|tools|>	• • • •	A tool's name and its details
<|plan|>	• • • •	The tools that are on
<|call|>	• • • •	One sentence: what it will do
<|result|>	• • • •	The reply to the person
<|answer|>	• • • •	What the tool returned, added by the app

3 **Put the app's loop in order. Here model() is a scripted stand-in that returns what a model would write. The loop must stop at an answer before running any tool, and print the answer at the end.**

Number the lines in order (1, 2, 3...), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `print(part)`

☐ `result = model(part)`

☐ `while True:`

☐ `part = model(text)`

☐ `if is_answer(part):`

☐ `text = text + part + result`

☐ `break`

☐ `text = request`

☐ `result = run_tool(part)`

4 **Which parts of an agent's text does the model write? Choose all that apply.**

☐ The tool call ☐ The tool's result ☐ The tools line ☐ The plan ☐ The answer

5 **This loop is meant to stop after five calls, but it never stops if the model keeps calling tools. Which line is the mistake?**

Circle the line with the mistake. Write it correctly:

```
1  text = request
2  steps = 0
3  while steps < 5:
4      steps = 0
5      part = model(text)
6      if is_answer(part):
7          break
8      text = text + part + run_tool(part)
```

✎ **Think of a task you might give an agent at school. Where in its loop would you want the app to stop and check, or ask you?**

Check your answers

Inside the loop

1 **A step limit that stops the loop and tells a person**

A step limit stops a loop that would run forever, and hands the problem to a person.

2 **<|tools|> — The tools that are on**
<|plan|> — One sentence: what it will do
<|call|> — A tool's name and its details
<|result|> — What the tool returned, added by the app
<|answer|> — The reply to the person

The tools line comes first, then the request; the model writes a plan, then a call; the app adds the result after the call marker; the answer ends the turn.

3 **Top to bottom: 8, X, 2, 3, 4, 7, 5, 1, 6**

```
text = request
while True:
    part = model(text)
    if is_answer(part):
        break
    result = run_tool(part)
    text = text + part + result
print(part)
```

The model writes the next part first. If it is the answer, the loop stops before any tool runs.

4 **The plan; The tool call; The answer**

The model writes the plan, the call and the answer.

5 **Line 4: `steps = steps + 1`**

Line 4 sets the counter back to 0 on every turn, so steps < 5 is always true and the loop can run forever.